

Vedant Sanjay Jadhav

vedantjadhav1414@gmail.com +91-7410036328
B.Tech. Computer Science (AI & ML), Pimpri Chinchwad University,
Pune

vedantjadhav-ai.vercel.app
github.com/VedantJadhav701 (34+ AI/ML
repos)
linkedin.com/in/vedantjadhav-ai ORCID

Summary

AI Engineer specializing in **Generative AI**, **Large Language Models (LLMs)**, **Vision Language Models (VLMs)**, **Domain-Specific Language Models (DSLMs)**, **Agentic AI**, and **Physical AI**. Experienced in building enterprise AI systems including RAG pipelines, multi-agent architectures, multimodal models, and efficient LLM inference using LangGraph, FastAPI, Qdrant, Hugging Face, PyTorch, and Docker. Published researcher (Scopus-indexed, Best Paper Award) with research focused on multi-agent systems, state-space models, healthcare AI, and embodied intelligence. Passionate about developing scalable AI systems from research to production.

Education

B.Tech. (AI & ML)	Pimpri Chinchwad University, Pune	8.10 CGPA	2023–27
Senior Secondary	HSC Board		2023
Secondary	SSC Board		2021

Experience

Tech Mahindra Jun. 2026 – Present
AI Engineer Intern Pune, Maharashtra

- Contributing to research and development of **Vision Language Models (VLMs)**, **Domain-Specific Language Models (DSLMs)**, and **Generative AI** systems for enterprise applications.
- Developing AI pipelines involving **Retrieval-Augmented Generation (RAG)**, semantic search, agentic AI workflows, and multimodal learning using modern open-source LLMs.
- Working with technologies including **Python**, **PyTorch**, **FastAPI**, **LangGraph**, **Hugging Face**, and **Ollama** to build scalable AI solutions and research prototypes.

DPulseAI Pvt. Ltd. Oct. 2025 – Apr. 2026
AI Engineer Intern Pune, Maharashtra

- Built production Generative AI and RAG systems; achieved **35% retrieval accuracy improvement** and **45% latency reduction** through hybrid chunking and reranking strategies
- Architected Dockerized LLM inference pipelines with CI/CD and canary deployments; reduced deployment time to **under 10 minutes** using GitHub Actions and Docker Registry
- Implemented KV cache quantization-based cost optimization and post-deployment drift monitoring; cut inference memory usage significantly on consumer-grade hardware

Projects

PharmaAssist DSLM GitHub

- Developed an enterprise-grade **Healthcare Domain-Specific Language Model (DSLIM)** leveraging **Retrieval-Augmented Generation (RAG)**, **LangGraph**, and **Qdrant** for citation-aware pharmaceutical knowledge retrieval.
- Designed a modular **agentic AI architecture** supporting semantic search, hybrid retrieval, medical document ingestion, and scalable healthcare AI workflows using lightweight open-source LLMs.

TinyVLM – Lightweight Vision Language Model GitHub

- Built a lightweight **Vision Language Model (VLM)** by integrating a vision encoder with a compact language model for multimodal understanding, image captioning, and visual question answering.
- Developed the complete multimodal inference pipeline including image preprocessing, feature projection, cross-modal alignment, and instruction-following generation using **PyTorch** and **Hugging Face**.

Research Jarvis – Agentic AI Research Assistant GitHub

- Developed an autonomous research assistant integrating **Hybrid RAG**, **LangGraph**, real-time arXiv monitoring, semantic search, and multi-agent orchestration for technical literature exploration.
- Built an end-to-end research pipeline with local document ingestion, citation-aware retrieval, FastAPI backend, and ChromaDB vector storage, enabling multi-hop reasoning over a large technical document corpus.

MAMA – Memory Augmented Mamba Agent GitHub

- Designed a modular autonomous AI architecture combining **Mamba-based reasoning**, hierarchical memory, LangGraph orchestration, and Retrieval-Augmented Generation for long-horizon reasoning tasks.
- Implemented memory management, tool calling, planning, and autonomous agent workflows supporting efficient local AI systems on consumer hardware.

Physical AI Robot Brain – Action Prediction System GitHub

- Developed a Physical AI pipeline using the **NVIDIA Physical AI dataset** to predict robot actions from sequential observations with a 3-layer LSTM implemented in PyTorch.
- Built an end-to-end robotics inference pipeline with FastAPI deployment, enabling autonomous action prediction for embodied AI and robotic decision-making systems.

LLM KV Cache Optimizer GitHub

- Developed a modular **KV Cache Quantization** framework reducing LLM memory footprint by **up to 23×** on consumer hardware while enabling efficient inference for 7B+ parameter models.
- Implemented INT8/INT4 quantization with dynamic precision switching and benchmarked performance across **LLaMA**, **Mistral**, and **Qwen** model families.

- Vision System – Autonomous Cleaning Robot

GitHub

 - Trained a custom **YOLOv8 Nano** model for real-world debris detection achieving **Precision 0.782, Recall 0.795, mAP@50 0.823** with inference at **~6 ms/image (~166 FPS)** on GPU edge hardware.
 - Built the complete computer vision pipeline from dataset collection and annotation to optimized deployment on an autonomous cleaning robot.

Research & Publications

- [Published, IJEMS 2025 – Scopus] *Ensemble and Hybrid ML Approaches for Renewable Energy Forecasting* – **Best Paper Award**, ICCTVB-25, Sanjay Ghodawat University – DOI
- [Preprint, 2026] *PhysicalMamba: State Space Models for Physical Planning and Robotic Task Execution* – DOI
- [Preprint, 2026] *RecursiveMAS: A Recursive Multi-Agent Latent Coordination Framework for Embodied Task Optimization* – DOI
- [Preprint, 2026] *ML Models to Predict Levulinic Acid Production from Sugarcane Bagasse* – DOI
- [Under Review] *Small Language Models for Clinical Reasoning and Medical Decision Alignment* – LLaMA, BioMistral, Med42v2, Qwen

Technical Skills

- Languages:** Python, C/C++
- Generative AI & Foundation Models:** Large Language Models (LLaMA, Qwen, Mistral), Vision Language Models (VLMs), Domain-Specific Language Models (DSLMs), Small Language Models (SLMs), Prompt Engineering, Fine-Tuning (LoRA/QLoRA), Retrieval-Augmented Generation (RAG), Agentic AI, Hybrid Search, KV Cache Quantization, Inference Optimization
- AI Frameworks & Orchestration:** LangChain, LangGraph, Hugging Face Transformers, Ollama, MCP (Model Context Protocol), CrewAI, Multi-Agent Systems, Tool Calling
- Machine Learning & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, Computer Vision (YOLOv8), NLP, CNNs, RNNs, LSTMs, Transformers, Mamba State Space Models
- Backend & MLOps:** FastAPI, Streamlit, Docker, MLflow, GitHub Actions (CI/CD), LLMOps, Model Serving, Drift Monitoring, Canary Deployment
- Vector Databases & Data:** Qdrant, ChromaDB, FAISS, PostgreSQL, MongoDB, Pandas, NumPy
- Research Areas:** Generative AI, Agentic AI, Healthcare AI, Physical AI, Embodied AI, Memory-Augmented AI, Multimodal AI, Autonomous AI Systems

Certifications

- | | |
|--|------|
| Deep Learning Specialization – DeepLearning.AI (Coursera) | 2025 |
| Supervised Machine Learning – Stanford University (Coursera) | 2025 |
| Mathematics for ML & Data Science – DeepLearning.AI (Coursera) | 2025 |

Achievements

- | | |
|---|------|
| 1st Place – Code4Society Hackathon 2026 (EcoGuard AI Carbon Platform) | 2026 |
| 1st Place – CodeApex 24-hour Hackathon, VIT | 2025 |
| Best Research Paper Award – ICCTVB-25, Sanjay Ghodawat University (Scopus Indexed) | 2025 |
| 3rd Place – National DevCraft Hackathon (Team Tech Titans), Fluxus 2025, IIT Indore | 2025 |
| 3rd Place – National AI Hackathon, IIT Indore | 2025 |
| National Finalist – Table Tennis, Atharv Ranbhoomi, IIM Indore | 2024 |